Jean Penny
Queensland Conservatorium,
Griffith University
mjpenny@bigpond.net.au

Amplified breath - (dis)embodied habitat: exploring sonic interfaces of performance, electronics, space and flautist identity

Abstract

The electronic spatialization of flute sound represents an intersection of acoustic resonance and technology, ancient sonic imagining and present day cultural imprints. The performative impacts of technological interventions and the new responses thus stimulated in performance practice, invite rigorous exploration and reflection. In this paper I aim to articulate my personal responses to spatialization technologies and live interactive electronics, to explore the expanded capacities provided for the flautist, and the new performative elements introduced by this genre. Focusing on breath tone and gesture, transformations of sonority, performer identity, space, physicality and interactivity will be investigated from the flautist's point of view. Warren Burt's new work for flute and live interactive electronics, Mantrae *(2007), is presented as an illumination of performance exploration. In this work, the movements of the flautist trigger sound events that radically alter the flute timbre, using camera signals to activate computer sound modification software, Plogue Bidule and Cycling '74's Hipno.*

Introduction

Solo flute works characteristically exploit spatial tonal qualities. The flute's resonance has motivated (and perhaps seduced) composers and performers to explore ways of capturing and expanding its translucent sonic qualities into evocative expression. This sonic ethos stretches across the centuries, from the architectural structures of Baroque works, such as J. S. Bach's *Partita in A minor* or Jacob van Eyck's *Der Fluyten Lust-hof*, to the more recent works of composers such as Takemitsu and Saariaho, in which the resonance and malleability of the tone and breath are a critical element of composition. Edgard Varese's seminal composition for solo flute, *Density 21.5* (1936), initiated new tonal directions through the use of key clicks, dynamics as sound manipulation and extremes of tessitura. This work generated immense change in the perception of flute sound and capabilities, and encouraged practitioners to explore new sonic paths. New texts began to appear examining such techniques as microtones, whistletones, breath tones, tongue clicks, key slaps, multiphonics, harmonic, glissandi, new embouchure techniques, vocalization and varieties of articulation by such authors as Bruno Bartoluzzi (Bartoluzzi, 1982), Pierre-Yves Artaud (Artaud, 1995) and Robert Dick (Dick, 1989). These techniques became the source material of composers, scrutinized for sonic properties and expanded out into the musical milieu through amplification and spatial projection.

Contemporary flute players have embraced these techniques, especially in conjunction with electronic interventions, revelling in the expanded sonic palette, a fresh abundance of ideas and new discoveries in expression and performance. The overlapping layers of instrumental performance and extended practice through electronic techniques have evolved into a complex search for interpretative cohesion and confluence, an engrossing search full of chance meetings with thoughts and new responses to movement, emphasis and projection.

My focus in this paper is on transformations of sonority, physicality, identity and space as they occur in the performance of music for flute and electronics. My vehicles for exploration are the breath tone, gesture and interactivated space.

Transformed Sonority

Amplification is the core electronic process that provides the technological means by which sonic transformations can be projected. With the convergence of extended flute techniques with amplification, the sonic landscape of flute music has been transformed, the inaudible has become audible, the interior has become exterior, and performer explorations of identity and expression aroused.

Breath tone begins with an unfocussed flute sound, which develops a character and strength of its own, an individual set of colours that can be manipulated acoustically and electronically to produce intense musical effects. Varied embouchure shape, internal mouth and throat shape, air stream direction, air speed, tongue positions, articulations, differing focus, use of vibrato, pitching, closed or open embouchure and body positions are used by the flautist to create the effects. These air sounds can be expanded into enormous wind noise, melded with

normal flute tone, voice, percussive techniques, greater or lesser amounts of hissing, or create tentative wisps or dots of micro sound. Percussive sounds on flute are usually key slaps or tongue clicks, and they are effectively used in combination with breath tones. A most astonishing soundscape results from extensive use of amplified percussive sounds, as seen in the third movement of Marco Stroppa's *little i* for flutes and electronics: *Moderato Percusive - Molto Minuzioso*. (Figure 1)
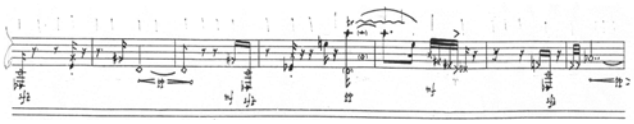


Figure 1, percussive articulations and breath tones in Stroppa's little i, third movement, bars 20 – 26.

In this music, various combinations of pizzicato, flutter tonguing, tongue ram, key slaps and breath tone are sent through speakers away from the player, generating an illusion of invisible presences and a sense of mystery and uncertainty. Diffusion techniques such as these, manipulations of timbre and sound placement, capturing and conversion techniques, expansion and contraction of the sound, all transform the breath tone into a viable and highly expressive musical device.

Hyper-magnification also generates an increased capacity to draw the listener into the intimate sound world of the performer. These expansions and contractions invoke changed expectations, impose new meanings and alter perceptions of the listener and performer. The breath tone sound can be stunningly large, creating a broad spatial implication, or coloured by minute tongue or percussion sounds, expanded out into the room, creating an impression of intimacy. Added vocalizations can add the sense of other presences, and the electronic processes such as delay or harmonization collect layers or mirroring reverberations to increase density or texture. These sounds can traverse across the room, or dart from speaker to speaker, create an immersion, or disconnection, or further illusions of intimacy or distance through panning or delay. The proximity of the breath tone can give an illusion to the audience of moving into the performer's sound space, an access to a private space, which can approach the confronting. The indistinct, grainy sound of breath tones can also produce a sense of uncertainty, of muffled undercurrent in the music, and can create a breathless, tense and dissonant sensation in the listener.

Working up the technique for breath tones is a remarkable journey, full of self-scrutinization and surprise. The very intimacy of the use of breath as a direct sound source places this technique in a different plane from normal resonant tone. Hearing this sound projected out into the room, shaping it and colouring it through minute mouth and blowing movements encourages much play and experimentation, generating a different physical approach, a re-positioning of the soloist in the space and new sense of identity.

Identity

The performer's identity is a combination of multiple interior and exterior factors. The outward identity (style, sound, interface) and inward identity (self awareness, interpretative imaginings) blend to form the performing persona. This persona is a projection of the self, intermingled with compositions and artistic goals. The basis of the instrumentalist's life is the commitment to music as a core activity. One's identity is strongly bound up with being a particular type of musician, and making specific musical sounds. As an extension of the self, the instrument acts as an amplifier. Inward and outward identity is displayed through performance, revealing aspects of inner processes and thoughts, responses and emotional connections, through interpretational signals. This self-projection creates a perception in both player and listener, imprinting both visual and aural impressions on a listener, and a sense of positioning and potential to the player.

A new sense of oneness in the sonic world can occur through electronic processes. These include the melding of techniques to give enhanced freedom of expression, sonic empowerment and flexibility. There is an expanded sense of the energy source – the gestures, breath, interpretations; and increased sense of sonic choice. There is a new sense of ensemble, and (at times) a spatialization of solo playing into layers and illusions of other presences. There is a magnification of the micro, intimate elements of playing, and the possibility to mask and confuse.

Timbre manipulations influence sonic identity, as new sound objects become familiar in one's sound world. Through memory, sounds become recognizable, "admissable" (Jonathan Harvey, 1999) and hence usable – and audiences also become significantly more comfortable with sounds they know or can relate to the known. The new sound worlds become elements you hold as part of yourself, part of your own "vocabulary", a representation of personality in sound. The map below (Figure 2) shows exploratory pathways as they pertain to breath tones in flute playing. The musical example included in the map is from Stroppa's *little i*, first movement, indicating a breath tone multiphonic changing to normal tone. Techniques are defined in the upper section of the map, and effects in the lower.

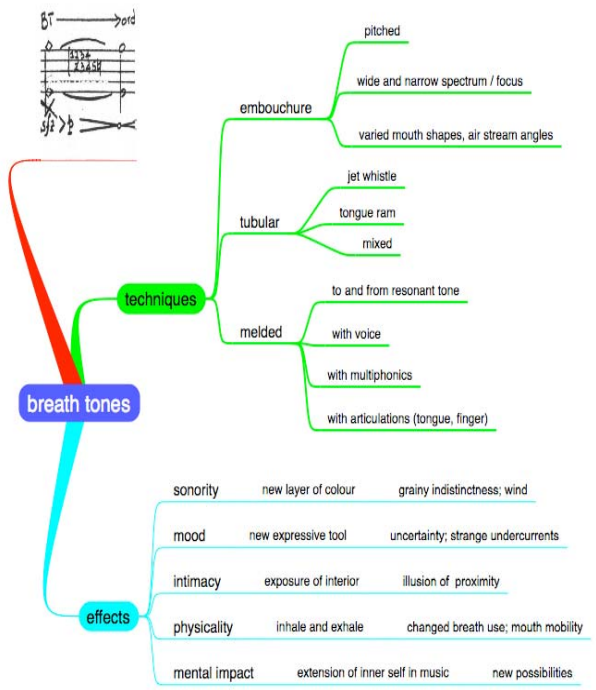


Figure 2, breath tone explorations

The following maps (Figures 3 and 4) indicate interior and exterior identity elements I have found in my own work, a representation of explorations and the pathways to new self-perception and awareness, projection and capabilities invoked by electronics.

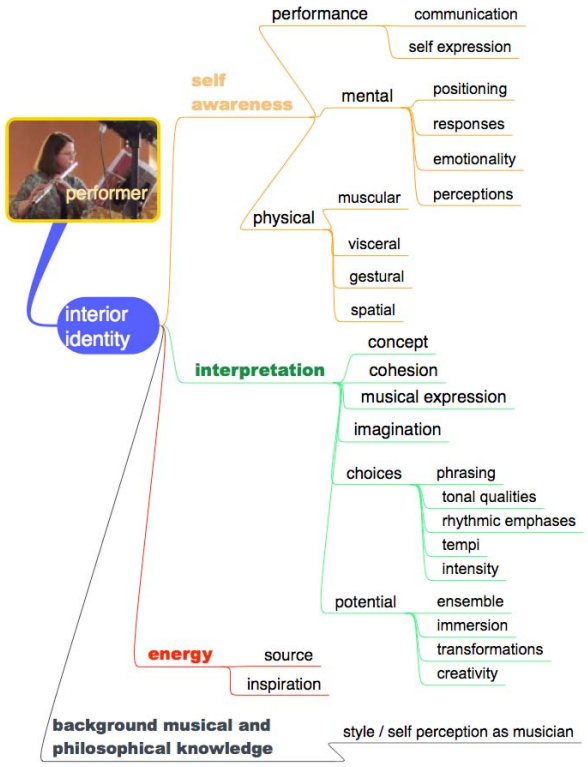


Figure 3, Performer identity explorations: interior

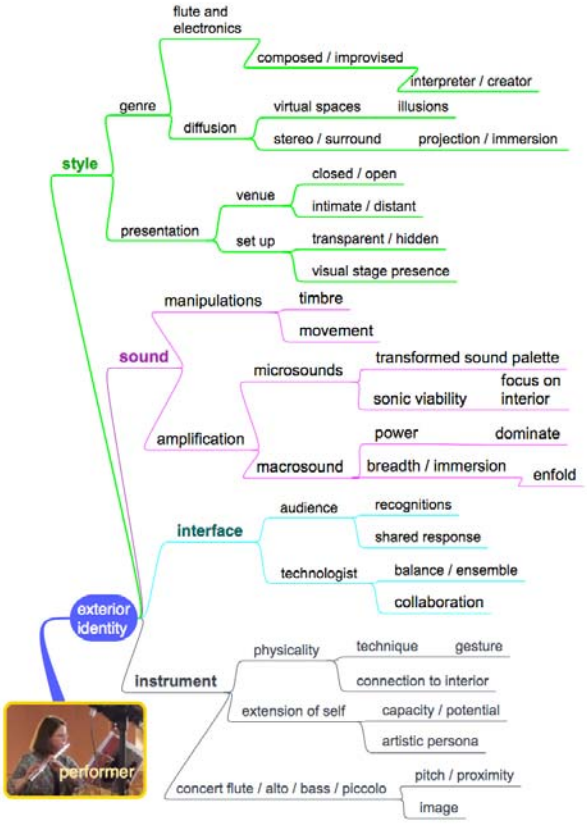


Figure 4, Performer identity explorations: exterior

Spaces: performer's habitat

> *The play–as-text can be performed in a space, but the play-as-event belongs to the space, and makes the space perform as much as the actors perform.* (Wiles, 2003 p.1)

The transformation of space in a performance for flute and electronics occurs with the physical set up and visible dynamics, the real space, and the virtual space of sonic illusion and invisible interconnections and processes. I alluded earlier to the sense of proximity created by hyper-magnification of intimate breath and tongue sounds. This expansion of internal bodily actions out into the room has a two-fold impact. The performer experiences an exposure of self, an open sharing of the internal in pursuit of sonic and communicative pathways. The self at the same time is empowered, enlarged, and given a new context for existence. The listener, on the other hand, experiences the extreme "close up", illusions and sensations of proximity and distance, unexpected sound movement and virtual dialogues with invisible sources.

The creation of a performance space is influenced by many elements of historical context or tradition and physical practicalities. Emergent modes of spatial representations, produced via technological, architectural and performative means have developed. The spaces of music (hall, church, outdoors), the forms of presenta-

tion (formal, intimate, invisible), the position of the audience (opposite, or around) all impact on the performance. The natural acoustic within the playing space has a major role in projection and synthesis, influencing listeners' and performers' perception of the music.

The influence of the sound in the space on the self in the space manifests in the way the player tunes in and 'plays' the space. Performing in a virtual space is entirely different to performing in an acoustic space. Similarities exist between surround sound and extremely reverberant cathedrals, for example, but the virtual spaces created through electronics do not necessarily follow natural rules of sonic behavior. The virtual space becomes a player in the performance, controlled in the main by the technologist. It is an element of expression and communication, and this changes the relationship of performer and performance. Sound sources can become entirely mixed up through live and pre-recorded sound sources, creating ambiguity, the sense of invisible presence and unexpected interconnections. The player can treat the space as a collaborator or an opposition. A sense of enclosure, or disorientation, uncertain proximity and passing of time can result, (for example, in Jean Claude Risset's *Passages*), or a dialogue with spatialized voices (as in Stroppa's *little i*). The performer can be magnified or obscured, empowered or reduced.

The dominant partnership creating the interconnections in this space is the instrumentalist – technologist relationship. This is an incontrovertibly vital, central element of electro-acoustic performance, but one that is rarely discussed from a performative point of view. The balance, the working methodology, the construction of a functional zone that activates and recreates the habitat for the piece are central elements to any performance. Putting together the hardware and the software, working together for confluence and functionality, frequently dominate preparation – the mechanics often demanding energy and attention throughout the rehearsal period. Re-interpretation of composed music implies a search for the composer's ideal, a pursuit of perfection. Premier performances are very likely experimental, an exploration of presentation ideas and a step towards this state of perfection. The relationship of instrumentalist and technologist is consequently one built on great trust, acceptance and a willingness to tread risky paths together. Most importantly, it also thrives when a commonality of artistic and aesthetic goals exist, when the myriad shades of balance, tonal softening or distortion, response and imagination create together a truly interactive gestural environment and ensemble.

Physicality and Gesture

> *Any sense of causality with visible instrumental gesture is secondary…..There is a strong tendency for the instrument to reestablish its presence separated from the electroacoustic soundworld. This is sometimes through body gestures….breath contour….but also…idiosyncracies of amplification.* (Emmerson (2007) p. 105)

Kinaesthetic awareness is heightened in performance with electronics through the intensification of focus on micro techniques, and the sense of physical freedom that comes from amplification. Flute playing is a mix of body and sound awareness. The emergence of sonic identity through gesture and shaping of musical material is founded on physical actions (posture, breathing, throat and mouth shaping, tonguing, arm and hand movement, emotionally reflective body language) learnt over many years of practice. Electronic processes impact very powerfully, highlighting variations of tone and intensity, and techniques that demand refined embouchure adjustments, new finger movements and microphone techniques. Amplification liberates the flautist's sense of energy: the sensation of projection is overwhelmingly overtaken by electronics, the audibility is no longer a matter for concern. The impact on the player is an increased relaxation of musculature, as the sound dimension is taken over, and subtle alterations can be made through minute movements. The focus is on this micro-sound world, on the link between physicality and sound texture and timbre, the projection of meaning in the interaction of breath, instrument and electronics.

The electronic separation of the sound from the body in the room gives the sound image an enlarged capacity, and invokes new listening and playing modes in the performer. Breath sounds, key movements and tonguing, are magnified and removed from the source, heard from a distance, giving these sonic units an entity and presence separate from the physical domain of the body. The breath tone is a fragile gesture, of blurred, diffuse and indefinite character. It is the link between interior and exterior elements, and represents a direct extension and magnification of the body inciting heightened perceptual awareness, focus on micro elements, and exploration of physical and expressive meanings and contexts. The immateriality of much of this sound world can also create a sense of the other, of dreams, and the separation of the sound from the body via electronics adds to this effect.

Articulation of breath tones employs physical gestures that take on meaning from electronic manipulations, musical context and perceptions of player and listener. For example, an explosive attack, or a soft pizzicato tongue sound will transmit opposite sensations; the emergence of breath tone from normal flute tone, or vice versa, will move the musical gesture towards a grainy, uncertain mood; the use of voice and breath tone will create impressions of multiple characters or shadows; percussive key slaps will add startling thwacks or pattering patterns; pitch movements will create mini skiffs of air, or sweeping gestures from one end of the flute to the other; jet whistles engage a more forceful muscular movement, and create in-

tensity and extroversion in the sound; tongue rams create a thud effect below the usual range of the flute. Ambiguity reigns in this environment, and a cognitive dissonance with reality can occur through the removal of visual gestural cues, the blurring of sound sources, and conjunction of virtual and real elements.

Interactivity

The first major work for flute and interactive electronics was Phillipe Manoury's *Jupiter* of 1987. This work developed and established score following with Max technologies in the field. Many works have followed, employing such triggering mechanisms as MIDI pedals, pitch sensors, movement sensors, and buttons with cables attached to the flute. The success of these processes can vary, depending at times on the characteristics of the performance venue. For example, Russell Pinkston's *Lizamander* for flute and Max/MSP (2003) uses pitch tracking. Two microphones are used, one for tracking and one for amplification and audio processing. The computer program detects specific pitches played by the flute at precise moments in the score, at times searching for notes above or below a particular threshold, in other instances, an exact pitch. This identification allows movement to the next section, triggering such actions as capturing and granulating sounds, harmonization, or the playing of computer-generated material, but does not always work effectively. If the hall is overly resonant, for example, the computer will have difficulty detecting pitches, and the work can only move forward with manual activation by the technologist.

An exploration of the use of breath as sonic material, and gesture as interactive trigger, introduces Warren Burt's new work, *Mantrae* (2007) for flute and live interactive electronics. Sharply articulated breath and flute tone provide the sonic base for this work, which captures the flautist's movements with a camera, and uses these to activate radical sound effects. The flautist moves between the three Mantrae at will, and the three music stands also, whilst maintaining a chant or speech-like style. The flute sound is played through a computer using Cycling '74's Hipno sound processing modules, and a random mixing routine is set up in Plogue Bidule, the host program for the processing. The soundscape is constantly changing, and the flautist is freely changing position and Mantra whilst maintaining a sense of focus and inner calm. The minimalist style and the conjunction of spiritual expression of the Mantrae further evoke ambiguity – a meditative calm in the midst of dramatic sound treatments, representing a constantly changing and unpredictable world. Kinaesthetic awareness is acute in the performance of this work, the broad gestural movements controlling sound events, serving as a challenge to the flautist to adopt a resolute, intense focus on the distilled yet powerful contour of the Mantrae, surrounded by complexity and multifariousness.

Conclusion

Through explorations of the interconnections of flute performance and technology, renewed sonic imaginings and responses build significant artistic discourse and illumination. Highlighting the multiple features of flute performance with technology, the transformation of the inaudible to the audible, the new approaches and perceptions of self and expression, new elements and new identities, new relationships and re-inventions of space and physical responses, the performer's world attains an expanded meaning and refreshed understanding. The instrument and performer become one, the microphone nourishes this relationship, and new sounds and identities are built.

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